

Edward Student

Lab 5 Consul

https://github.com/EdvardStudent23/apz_hw/tree/micro_consul

Запустив сервіси на докері:

```
kpoint snapshot min: 38, snapshot max: 38 snapshot count: 0, oldest timestamp: (0, 0), meta checkpoint timestamp: (0, 0) base write gen: 1"}}
```

counter-service-2		INFO:	172.24.0.6:52020	-	"GET /health HTTP/1.1"	200	OK
facade-service-1		INFO:	172.24.0.6:60778	-	"GET /health HTTP/1.1"	200	OK
facade-service-2		INFO:	172.24.0.6:55686	-	"GET /health HTTP/1.1"	200	OK
logging-3		INFO:	172.24.0.6:49204	-	"GET /health HTTP/1.1"	200	OK
logging-2		INFO:	172.24.0.6:38528	-	"GET /health HTTP/1.1"	200	OK
logging-1		INFO:	172.24.0.6:34532	-	"GET /health HTTP/1.1"	200	OK
counter-service-1		INFO:	172.24.0.6:53938	-	"GET /health HTTP/1.1"	200	OK
counter-service-2		INFO:	172.24.0.6:36924	-	"GET /health HTTP/1.1"	200	OK

Я запустив сервіси на консул:

The screenshot shows the Consul UI interface. The browser address bar indicates the URL is `localhost:8500/ui/dc1/services`. The left sidebar contains a navigation menu with the following items: Overview, Services (selected), Nodes, Key/Value, Intentions, Access Controls, Tokens, Policies, Roles, Auth Methods, Organization, and Peers. The main content area is titled 'Services' and shows '4 total' services. Below the title is a search bar and a 'Search Across' dropdown. There are also filters for 'Health Status' and 'Service Type', and a sort button labeled 'Unhealthy to Healthy'. The list of services is as follows:

Service	Health	Instances	Tags
consul	Healthy	1 instance	
counter-service	Healthy	2 instances	counter-service, fastapi, python
facade-service	Healthy	2 instances	facade-service, fastapi, python
logging-service	Healthy	3 instances	fastapi, logging-service, python

< All Services

counter-service

Instances Intentions Routing Tags

Q Search

Search Across ▾

Health Status ▾

↕ Unhealthy to Healthy ▾

counter-service-79988b27

✓ All service checks passing ✓ All node checks passing ➦ b7c53367e346 🌐 counter-service-1:8000 📄 cour

counter-service-e37c7398

✓ All service checks passing ✓ All node checks passing ➦ b7c53367e346 🌐 counter-service-2:8000 📄 cour

< All Services

facade-service

Instances Intentions Routing Tags

Q Search

Search Across ▾

Health Status ▾

↕ Unhealthy to Healthy ▾

facade-service-3ceb0a18

✓ All service checks passing ✓ All node checks passing ➦ b7c53367e346 🌐 facade-service-2:8000 📄 facad

facade-service-93508d67

✓ All service checks passing ✓ All node checks passing ➦ b7c53367e346 🌐 facade-service-1:8000 📄 facad

IP-адреси і порти зчитуються facade-service з Consul. В коді чи конфігураціях не задано статичних значень адрес:

```

y
INFO:      172.24.0.6:42830 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:47912 - "GET /health HTTP/1.1" 200 OK
2026-05-04 09:52:11,447 INFO [Consul] Resolved 'logging-service' -> http://lo
gging-1:8000 (from 3 instances)
2026-05-04 09:52:11,586 INFO HTTP Request: POST http://logging-1:8000/log "HT
TP/1.1 200 OK"
2026-05-04 09:52:11,598 INFO [facade-service @ http://facade-service-1:8000]
enqueued 0b3f16e8-f9fa-4a71-a8e8-755f05033013
INFO:      172.24.0.1:59222 - "POST /process HTTP/1.1" 200 OK
INFO:      172.24.0.6:54788 - "GET /health HTTP/1.1" 200 OK
2026-05-04 09:52:13,567 INFO [Consul] Resolved 'logging-service' -> http://lo
gging-1:8000 (from 3 instances)
2026-05-04 09:52:13,587 INFO HTTP Request: POST http://logging-1:8000/log "HT
TP/1.1 200 OK"
2026-05-04 09:52:13,589 INFO [facade-service @ http://facade-service-1:8000]
enqueued b1e4bb94-b6c6-4ab9-880e-f94bff4942d1
INFO:      172.24.0.1:59232 - "POST /process HTTP/1.1" 200 OK
2026-05-04 09:52:15,539 INFO [Consul] Resolved 'logging-service' -> http://lo
gging-3:8000 (from 3 instances)
2026-05-04 09:52:15,600 INFO HTTP Request: POST http://logging-3:8000/log "HT

```

```

a787-460b-b957-9ee2308eafc4", "status": "queued"}
PS C:\Users\user\Documents\3_course\sem2\apz\hw> docker logs facade-service-1
--tail=30
INFO:      172.24.0.6:46776 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:39854 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:57822 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:44248 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:44786 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:57202 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:50306 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:44678 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:55902 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:44204 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:52726 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:55664 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:49354 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:36652 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:53086 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:47172 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:57800 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:57776 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:45182 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:34182 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:38950 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:47100 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:47680 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:50438 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:52452 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:48624 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:51984 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:55728 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:37552 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:38626 - "GET /health HTTP/1.1" 200 OK
PS C:\Users\user\Documents\3_course\sem2\apz\hw>

```

facade-service не має статичної адреси. Він щоразу запитує Consul: де logging-service?

Починаю надсилати транзакції

< All Services

logging-service

Instances Intentions Routing Tags

Search Search Across

Health Status Unhealthy to Healthy

logging-service-640dedd2
✓ All service checks passing ✓ All node checks passing b7c53367e346 logging-3:8000 logging-sen

logging-service-a917b677
✓ All service checks passing ✓ All node checks passing b7c53367e346 logging-1:8000 logging-sen

logging-service-e0f4d124
✓ All service checks passing ✓ All node checks passing b7c53367e346 logging-2:8000 logging-sen

Перевіряю Key/Value:

config Create

Search

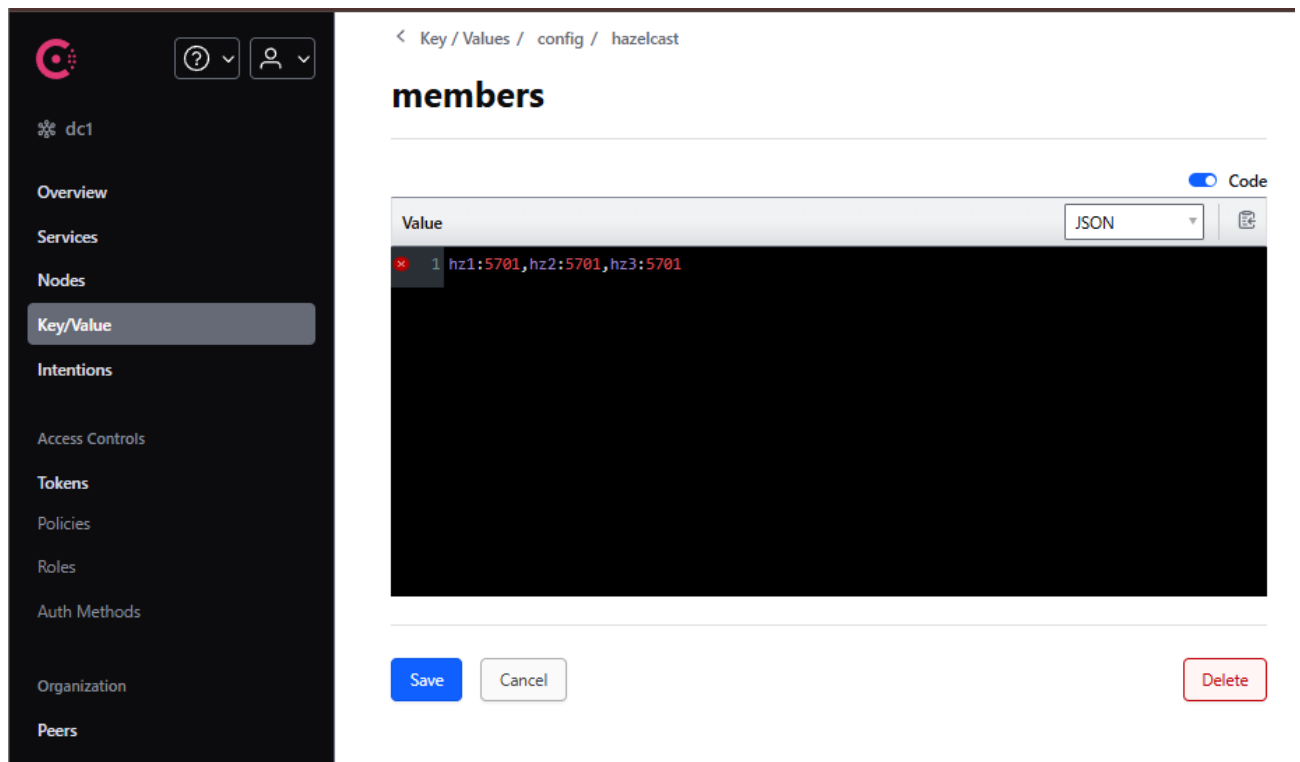
Type Folders to Keys

Name	Actions
hazelcast	...
mq	...

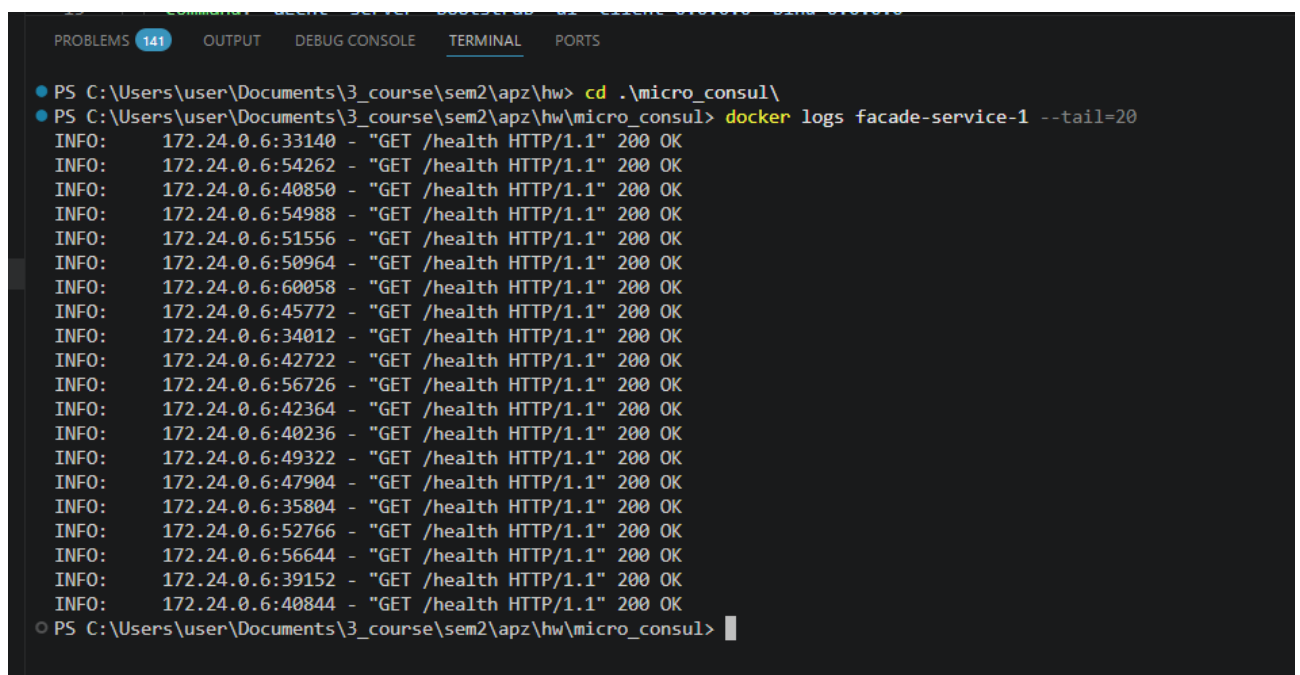
Key/Value
ACLs are not currently enabled in this cluster

Access Controls
Tokens

Відкриваю наприклад hazelcast/members:



Отже, Hazelcast і MQ конфіги зберігаються в Consul як key/value (завдання 3 і 4)



Вміст Key/Value для config/hazelcast/

```
● PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> curl http://localhost:8500/v1/kv/config/hazelcast/members?raw
```

Security Warning: Script Execution Risk

Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.

RECOMMENDED ACTION:

Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help
(default is "N"):Y

```
Statuscode      : 200
StatusDescription : OK
Content          : hz1:5701,hz2:5701,hz3:5701
RawContent       : HTTP/1.1 200 OK
                  Content-Security-Policy: sandbox
                  Vary: Accept-Encoding
                  X-Consul-Default-Acl-Policy: allow
                  X-Consul-Index: 18
                  X-Consul-Knownleader: true
                  X-Consul-Lastcontact: 0
                  X-Consul-Query-Ba...
Forms            : {}
Headers          : {[Content-Security-Policy, sandbox], [Vary, Accept-Encoding], [X-Consul-Default-Acl-Policy, allow], [X-Consul-Index, 18]...}
Images           : {}
InputFields      : {}
Links            : {}
ParsedHtml       : mshtml.HTMLDocumentClass
RawContentLength : 26
```

```
● PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> █
```

```
curl http://localhost:8500/v1/kv/config/hazelcast/cluster-name?rawro_consul>

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web
page might be run when the page is parsed.
RECOMMENDED ACTION:
    Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help
(default is "N"):Y

StatusCode      : 200
StatusDescription : OK
Content         : dev
RawContent      : HTTP/1.1 200 OK
                  Content-Security-Policy: sandbox
                  Vary: Accept-Encoding
                  X-Consul-Default-Acl-Policy: allow
                  X-Consul-Index: 19
                  X-Consul-Knownleader: true
                  X-Consul-Lastcontact: 0
                  X-Consul-Query-Ba...
Forms           : {}
Headers         : {[Content-Security-Policy, sandbox], [Vary, Accept-Encod
                  ing], [X-Consul-Default-Acl-Policy, allow], [X-Consul-In
                  dex, 19]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 3

PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul>
```

```
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help
(default is "N"):Y

StatusCode      : 200
StatusDescription : OK
Content         : logs_map
RawContent      : HTTP/1.1 200 OK
                  Content-Security-Policy: sandbox
                  Vary: Accept-Encoding
                  X-Consul-Default-Acl-Policy: allow
                  X-Consul-Index: 20
                  X-Consul-Knownleader: true
                  X-Consul-Lastcontact: 0
                  X-Consul-Query-Ba...

Forms          : {}
Headers        : {[Content-Security-Policy, sandbox], [Vary, Accept-Encod
                  ing], [X-Consul-Default-Acl-Policy, allow], [X-Consul-In
                  dex, 20]...}

Images         : {}
InputFields    : {}
Links          : {}
ParsedHtml     : mshtml.HTMLDocumentClass
RawContentLength : 8

PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul>
```

Логи:

```
INFO: Started server process [1]
INFO: Waiting for application startup.
2026-05-04 09:48:41,345 INFO [Consul] Registered 'logging-service' as 'loggin
g-service-9802bf4c' at logging-1:8000
2026-05-04 09:48:41,345 INFO [logging-service @ http://logging-1:8000] regist
ered in Consul
2026-05-04 09:48:41,361 INFO [Consul KV] config/hazelcast/members = 'hz1:5701
,hz2:5701,hz3:5701'
2026-05-04 09:48:41,367 INFO [Consul KV] config/hazelcast/cluster-name = 'dev
'
2026-05-04 09:48:41,378 INFO [Consul KV] config/hazelcast/map-name = 'logs_ma
p'
2026-05-04 09:48:41,378 INFO [logging-service] KV → cluster='dev' members=hz1
:5701,hz2:5701,hz3:5701 map='logs_map'
2026-05-04 09:48:41,380 INFO HazelcastClient 5.3.0 is STARTING
2026-05-04 09:48:41,380 INFO HazelcastClient 5.3.0 is STARTED
```

MQ configuration:

Error response from daemon: No such container: logging-service

● PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> curl http://localhost:8500/v1/kv/config/mq/queue-name?raw

Security Warning: Script Execution Risk

Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.

RECOMMENDED ACTION:

Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help
(default is "N"):Y

StatusCode : 200

StatusDescription : OK

Content : transactions_queue

RawContent : HTTP/1.1 200 OK

Content-Security-Policy: sandbox

Vary: Accept-Encoding

X-Consul-Default-Acl-Policy: allow

X-Consul-Index: 23

X-Consul-Knownleader: true

X-Consul-Lastcontact: 0

X-Consul-Query-Ba...

Forms : {}

Headers : {[Content-Security-Policy, sandbox], [Vary, Accept-Encoding], [X-Consul-Default-Acl-Policy, allow], [X-Consul-Index, 23]...}

Images : {}

InputFields : {}

Links : {}

ParsedHtml : mshtml.HTMLDocumentClass

RawContentLength : 18

○ PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul>

```

PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> curl http://localhost:8500/v1/kv/config/mq/members?raw

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
RECOMMENDED ACTION:
Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help
(default is "N"):Y

StatusCode      : 200
StatusDescription : OK
Content          : hz1:5701,hz2:5701,hz3:5701
RawContent       : HTTP/1.1 200 OK
                  Content-Security-Policy: sandbox
                  Vary: Accept-Encoding
                  X-Consul-Default-Acl-Policy: allow
                  X-Consul-Index: 21
                  X-Consul-Knownleader: true
                  X-Consul-Lastcontact: 0
                  X-Consul-Query-Ba...

Forms           : {}
Headers         : {[Content-Security-Policy, sandbox], [Vary, Accept-Encoding], [X-Consul-Default-Acl-Policy, allow], [X-Consul-Index, 21]...}

Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 26

PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul>

```

MQ конфігурація у Consul KV:

<http://localhost:8500> - Key/Value - config/mq/

mq

Create

Q Search

Type ▾

⇅ Folders to Keys ▾

Name	Actions
cluster-name	...
members	...
poll-timeout-sec	...
queue-name	...

cluster-name

☒ Code

Value

JSON

1 dev

Save

Cancel

Delete

< Key / Values / config / mq

members

Value

JSON

1

hz1:5701,hz2:5701,hz3:5701

Save

Cancel

Delete

< Key / Values / config / mq

queue-name

Value

JSON

1

transactions_queue

< Key / Values / config / mq

poll-timeout-sec

Value

JSON

1

2

```

PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> docker logs facade-service-1 --tail=20
INFO:      172.24.0.6:49756 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:45406 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:39544 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:53020 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:38860 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:37578 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:38452 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:41212 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:54502 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:34958 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:51844 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:50130 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:49622 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:52410 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:37900 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:34506 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:39920 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:60870 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:35508 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:38400 - "GET /health HTTP/1.1" 200 OK
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> docker logs counter-service-1 --tail=20
INFO:      172.24.0.6:58150 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:40016 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:36846 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:50896 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:45888 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:38484 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:43956 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:33224 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:44584 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:59770 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:60114 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:37934 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:56400 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:47170 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:38556 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:54304 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:33632 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:48976 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:41664 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:46134 - "GET /health HTTP/1.1" 200 OK
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul>

```

5. Відключення екземпляру відобразатиметься у Consul, а виклики перенаправлятимуться до інших екземплярів:

Термінал 1:

```
for ($i=1; $i -le 15; $i++) {
```

```

$body = @{
    user_id = "test"
    amount = 10
} | ConvertTo-Json

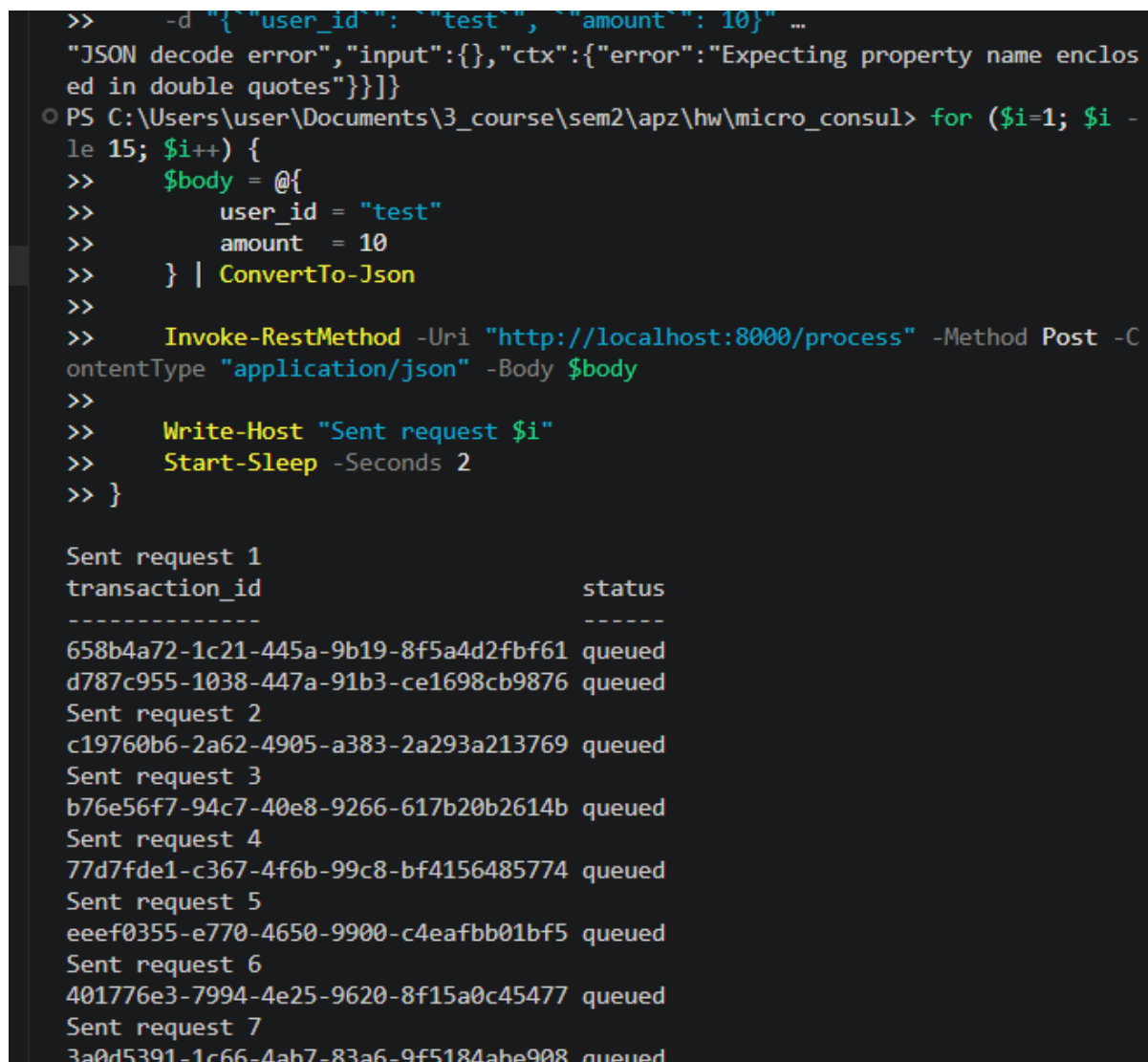
Invoke-RestMethod -Uri "http://localhost:8000/process" -Method Post -ContentType "application/json" -
Body $body

Write-Host "Sent request $i"

Start-Sleep -Seconds 2

}

```



```

>> -d '{"user_id": "test", "amount": 10}' ...
"JSON decode error","input":{},"ctx":{"error":"Expecting property name enclosed in double quotes"}}}]
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> for ($i=1; $i -le 15; $i++) {
>>     $body = @{
>>         user_id = "test"
>>         amount = 10
>>     } | ConvertTo-Json
>>
>>     Invoke-RestMethod -Uri "http://localhost:8000/process" -Method Post -ContentType "application/json" -Body $body
>>
>>     Write-Host "Sent request $i"
>>     Start-Sleep -Seconds 2
>> }

Sent request 1
transaction_id      status
-----
658b4a72-1c21-445a-9b19-8f5a4d2fbf61 queued
d787c955-1038-447a-91b3-ce1698cb9876 queued
Sent request 2
c19760b6-2a62-4905-a383-2a293a213769 queued
Sent request 3
b76e56f7-94c7-40e8-9266-617b20b2614b queued
Sent request 4
77d7fde1-c367-4f6b-99c8-bf4156485774 queued
Sent request 5
eeef0355-e770-4650-9900-c4eafbb01bf5 queued
Sent request 6
401776e3-7994-4e25-9620-8f15a0c45477 queued
Sent request 7
3a0d5391-1c66-4ab7-83a6-9f5184abe908 queued

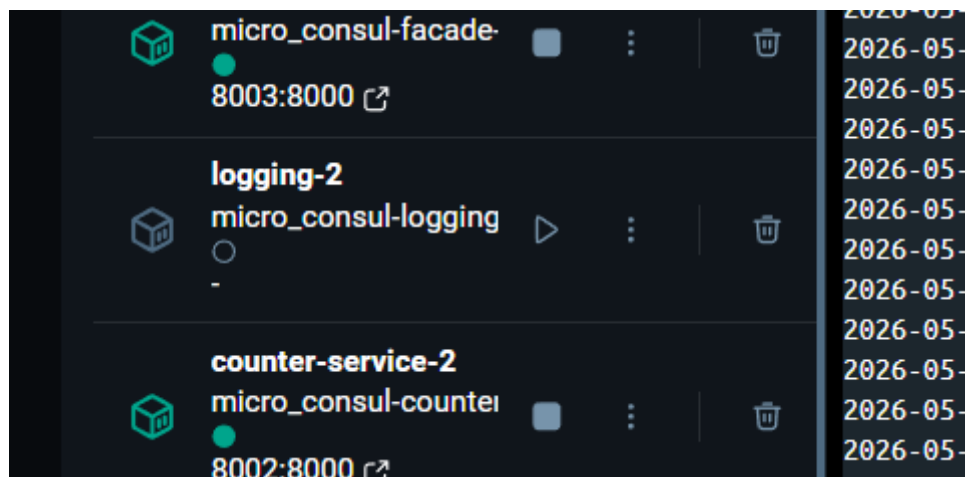
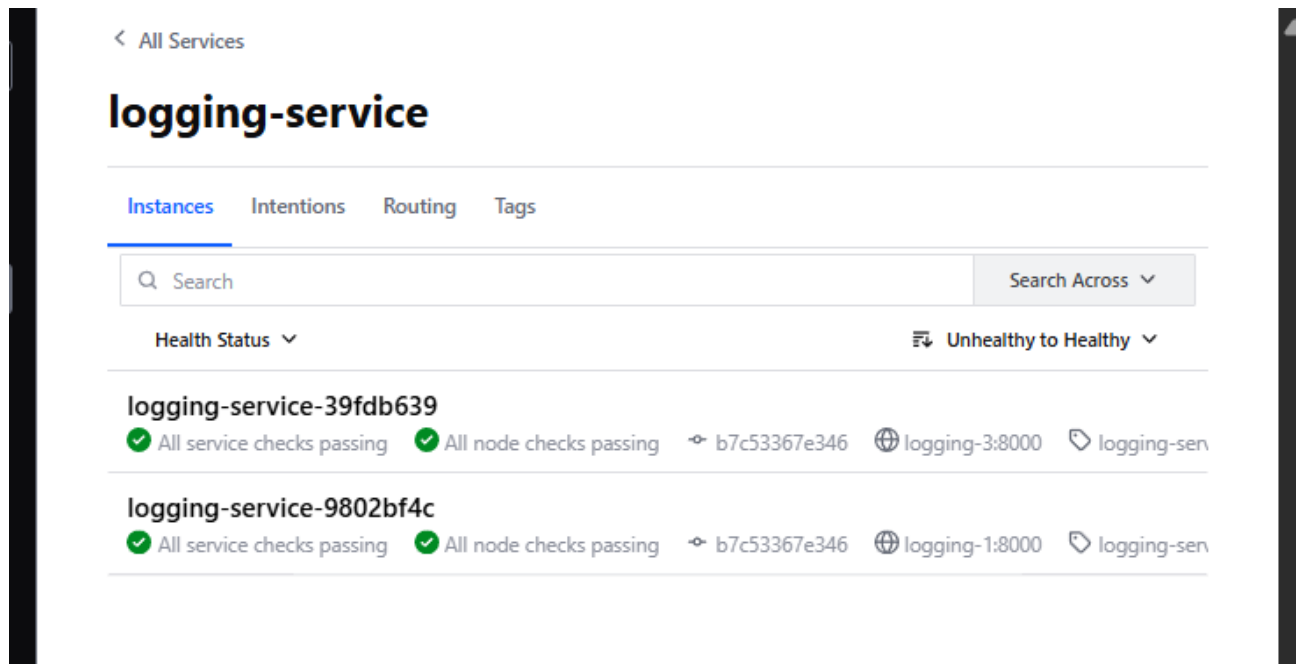
```

Запити почали надсилатися.

Термінал 2: зупиняю 1 екземпляр.

docker stop logging-2

Після цієї команди на logging-service зник 1 інстанс:



Service Discovery - система стійка до відмов. Consul постійно перевіряє стан кожного екземпляру через health check. Якщо екземпляр падає - Consul одразу прибирає його зі списку живих, і facade більше не надсилає запити на мертву адресу. Все це відбувається автоматично без перезапуску системи

Запити після відключення однієї ноди й далі йдуть. Почав ще раз надсилення і воно працює:

```
>>
>> Invoke-RestMethod -Uri "http://localhost:8000/process" -Method Post -
ontentType "application/json" -Body $body
>>
>> Write-Host "Sent request $i"
>> Start-Sleep -Seconds 2
>> }
```

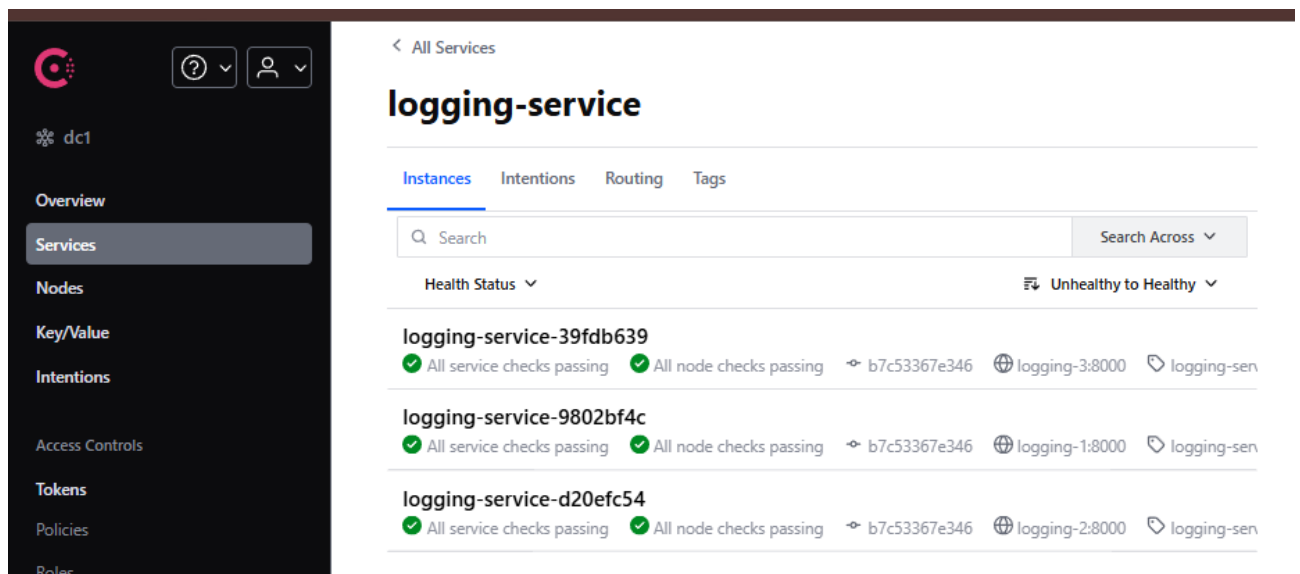


```
Sent request 1
transaction_id      status
-----
d42f0f89-572c-4194-bf85-6d77265a8e51 queued
1cb50afc-b0fc-4db4-b52c-873fb64f4eb6 queued
Sent request 2
5063cdfa-2b31-4fa4-b1fb-9dba349f2312 queued
Sent request 3
ccd74b9a-3042-4838-a75b-21f4194c87d9 queued
Sent request 4
464f5f49-e448-4de5-8fb8-a3d7a19cb756 queued
Sent request 5
```

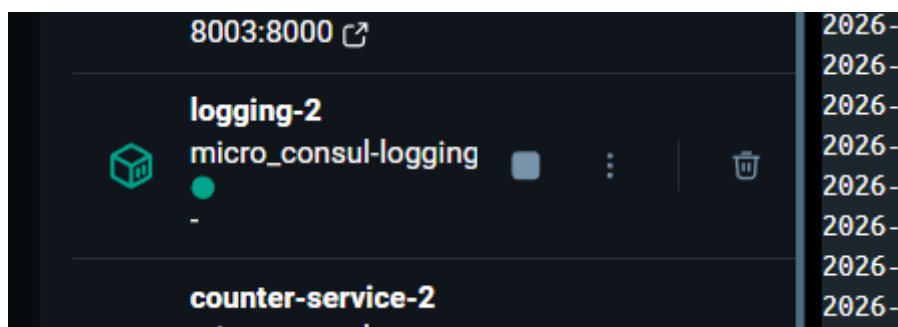
Запускаю ще раз логінг 2:

docker start logging-2:

На Consul UI я знову бачу 3 ноди:



Також ось піднятий сервіс на докері:



Task 6: скріншоти з POST/GET-запитами та відповіді на них

Надсилання POST запитів:

```
$body1 = @{ user_id = "alice"; amount = 500 } | ConvertTo-Json
```

```
Invoke-RestMethod -Uri "http://localhost:8000/process" -Method Post -ContentType "application/json" -  
Body $body1
```

```
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> $body1 = @{ use  
r_id = "alice"; amount = 500 } | ConvertTo-Json  
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> Invoke-RestMeth  
od -Uri "http://localhost:8000/process" -Method Post -ContentType "applicatio  
n/json" -Body $body1
```

transaction_id	status
90d8f027-2f15-4ad6-80cd-9db7b2d23f9d	queued

Другий запит:

```
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> $body2 = @{ use  
r_id = "bob"; amount = 300 } | ConvertTo-Json  
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> Invoke-RestMeth  
od -Uri "http://localhost:8000/process" -Method Post -ContentType "applicatio  
n/json" -Body $body2
```

transaction_id	status
360caef4-3721-4509-ae8f-7e4f0da3106a	queued

Перевірка результатів GET запитів:

Отримання лоогів усіх акаунтів:

```
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> Invoke-RestMeth  
od -Uri "http://localhost:8000/accounts"
```

test	alice	bob
750	500	300

Ось результат вичитаний з Hazelcast, MQ

Також ось логи конкретного користувача:

```

● PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> Invoke-RestMethod -Uri "http://localhost:8000/user/alice"

balance transactions
-----
500 {@{user_id=alice; amount=500; transaction_id=90d8f027-2f15-4ad6-8...

```

Успішно підтверджує що logging сервіс успішно зберіг дані в Hazelcast Map.

Task 7 – консолі сервісів:

Вміст консолі кожного з мікросервісів якщо туди виводиться якась корисна інформація

Facade:

```

context canceled
● PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> docker logs facade-service-1 --tail=30
INFO:      172.24.0.6:36090 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:50206 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:54862 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:35070 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:50712 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:36502 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:37908 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:43286 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:48546 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:55398 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:34984 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:46738 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:35540 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:40884 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:49644 - "GET /health HTTP/1.1" 200 OK

```

Counter:

```

INFO:      172.24.0.6:52672 - "GET /health HTTP/1.1" 200 OK
● PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> docker logs counter-service-1 --tail=30
INFO:      172.24.0.6:33474 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:46796 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:58036 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:47868 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:45370 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:49192 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:52572 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:51444 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:47420 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:53792 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:49068 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:59482 - "GET /health HTTP/1.1" 200 OK
INFO:      172.24.0.6:35692 - "GET /health HTTP/1.1" 200 OK

```

Logging:

```
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> docker logs log
ging-1 --tail=30
INFO: 172.24.0.6:47260 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:47836 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:59414 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:48226 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:54450 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:49386 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:57134 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:39882 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:37372 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:47540 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:50646 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:48580 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:44816 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:48280 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:54844 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:48228 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:38770 - "GET /health HTTP/1.1" 200 OK
INFO: 172.24.0.6:60840 - "GET /health HTTP/1.1" 200 OK
```

Протестував продуктивність на прикладі тестів з лаб 1 та лаб 3:

```
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul> python test_per
f_task5.py

Starting Test: 10 clients, 10000 reqs each (Same account: False)
Total Time: 2849.27s
Requests Per Second: 35.10
Logging Overhead: 12198.47s (428.1%)
Counter Overhead: 4303.19s (151.0%)

Starting Test: 10 clients, 10000 reqs each (Same account: True)
Total Time: 2250.41s
Requests Per Second: 44.44
Logging Overhead: 10172.90s (452.0%)
Counter Overhead: 3406.96s (151.4%)
PS C:\Users\user\Documents\3_course\sem2\apz\hw\micro_consul>
```

Ось загальна таблиця з результатами продуктивності для task 1, 3 та 5:

Test scenarios	Metric	Task 1 (in-mem)	Task 3 (DB)	Task 5 (final / Consul)
10 accounts	Total time	1413s	1644.41s	2849.27s

	logging-service contribution	5770s	6347.13s	12198.47s
	counter-service contribution	5821s	5844.44s	4303.19s
1 account	Total time	1331.92s	1611.52s	2250.41s
	logging-service contribution	5419.74s	6116.56s	10172.90s
	counter-service contribution	5511.33s	5885.38s	3406.96s

Task 5 повільніший за logging: при кожному POST-запиті facade звертається до Consul за адресою logging-service - це додатковий мережевий round-trip якого не було в Task 1 і Task 3. Через це logging contribution зріс майже вдвічі.

counter contribution знизився: counter тепер повністю асинхронний через Hazelcast Queue - facade просто кладе повідомлення в чергу і не чекає відповіді від counter. Тому його внесок у час facade менший.